

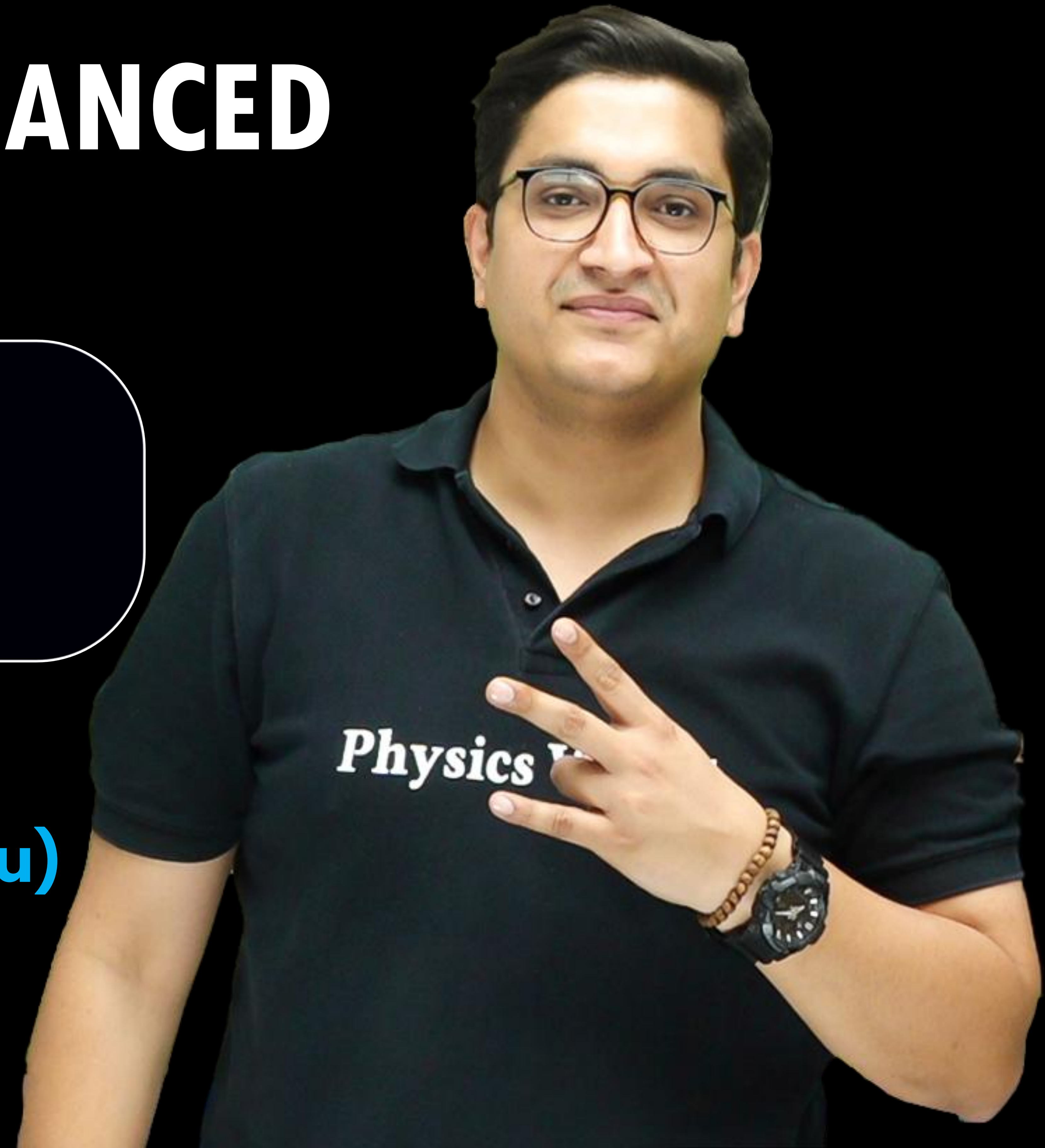
JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

LIMITS

DIBY (DO IT BY YOURSELF)

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DIBY-01



$$L = \lim_{x \rightarrow \infty} \frac{(1+2+3+\dots+n)(1^3+2^3+3^3+\dots+n^3)}{(1^2+2^2+3^2+\dots+n^2)^2}$$

[Ans. : $\frac{9}{8}$]

DIBY-02

The value of $\lim_{h \rightarrow 0} \left\{ \frac{\sqrt{3} \sin\left(\frac{\pi}{6} + h\right) - \cos\left(\frac{\pi}{6} + h\right)}{\sqrt{3}h(\sqrt{3} \cosh - \sinh)} \right\}$ is :

- (A) $\frac{3}{4}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{4}{3}$ (D) $\frac{2}{3}$

[JEE Mains-2021]

[Ans. : (D)]

DIBY-03

The value of $\lim_{x \rightarrow 0} \left(\frac{x}{\sqrt[8]{1-\sin x} - \sqrt[8]{1+\sin x}} \right)$ is :

- (A) 0 (B) 4 (C) -4 (D) -1

[JEE Mains-2021]

[Ans. : (C)]

DIBY-04

Let $f : R \rightarrow R$ be a function such that $f(2) = 4$ and $f'(2) = 1$. Then, the value of

$\lim_{x \rightarrow 2} \frac{x^2 f(2) - 4f(x)}{x-2}$ is equal to

[JEE Mains-2021]

- (A) 4 (B) 8 (C) 16 (D) 12

[Ans.: (D)]

DIBY-05

Let $a_1, a_2, a_3, \dots, a_n$ be n positive consecutive terms of an arithmetic progression.

If $d > 0$ is its common difference, then $\lim_{n \rightarrow \infty} \sqrt{\frac{d}{n}} \left(\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} \right)$ is:

[JEE Mains-2023]

- (A) 0 (B) $\sqrt{d}$ (C) 1 (D) $\frac{1}{\sqrt{d}}$

[Ans.: (C)]



DIBY-06

$\lim_{x \rightarrow \infty} \frac{(x+1)^{10} + (x+2)^{10} + \dots + (x+100)^{10}}{x^{10} + 10^{10}}$ is equal to:

[Ans. : 100]

DIBY-07

Find the value of $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+1} - \sqrt[3]{x^2+1}}{6x + \sqrt[4]{x^4+1} - \sqrt[5]{x^4+1}}$

[Ans. : -1/5]

DIBY-08

$\lim_{x \rightarrow \frac{1}{\sqrt{2}}} \frac{\sin(\cos^{-1} x) - x}{1 - \tan(\cos^{-1} x)}$ is equal to:

[JEE Mains-2022]

(A) $\sqrt{2}$

(B) $-\sqrt{2}$

(C) $\frac{1}{\sqrt{2}}$

(D) $-\frac{1}{\sqrt{2}}$

[Ans. : (D)]

DIBY-09

The value of $\lim_{x \rightarrow \frac{1}{\sqrt{2}}} \frac{x - \cos(\sin^{-1} x)}{1 - \tan(\sin^{-1} x)}$ is:

(A) $1/\sqrt{2}$

(B) $-1/\sqrt{2}$

(C) 1

(D) Does not exist
[Ans.: (B)]

[JEE Mains-2022]

DIBY-10

$\lim_{x \rightarrow 0^-} \frac{\sqrt{\pi} - \sqrt{2\sin^{-1} x}}{\sqrt{1-x}}$ is equal to:

(A) $\frac{1}{\sqrt{2\pi}}$

(B) $\sqrt{\frac{2}{\pi}}$

(C) $\sqrt{\frac{\pi}{2}}$

(D) $\sqrt{\pi}$

[Ans.: (B)]

[JEE Mains-2019]

DIBY-11

If $a = \lim_{x \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + x^4} - \sqrt{2}}}{x^4}$ and $b = \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}}$, then the value of ab^3 is

[JEE Mains-2024]

(A) 36

(B) 32

(C) 25

(D) 30

[Ans. : (B)]

DIBY-12

If $\lim_{x \rightarrow 0} \left\{ x^{\frac{1}{8}} \left(1 - \cos \frac{x^2}{2} - \cos \frac{x^2}{4} + \cos \frac{x^2}{2} \cos \frac{x^2}{4} \right) \right\} = 2^{-k}$, then the value of k is _____.

[JEE Mains-2020]

[Ans. : 8]

DIBY-13

$\lim_{x \rightarrow \frac{\pi}{4}} \frac{8\sqrt{2} - (\cos x + \sin x)^7}{\sqrt{2} - \sqrt{2} \sin 2x}$ is equal to:

[JEE Mains-2022]

(A) 14

(B) 7

(C) $14\sqrt{2}$

(D) $7\sqrt{2}$

[Ans. : (A)]

USE OF EXPANSION & FINDING UNKNOWN IF LIMIT IS GIVEN



DIBY-14

Find the values of a and b in order that $\lim_{x \rightarrow 0} \frac{x(1+a\cos x)-b\sin x}{x^3} = 1$

$$[\text{Ans. : } a = \frac{-5}{2}, b = \frac{-3}{2}]$$

DIBY-15

Find the integral value of n for which $\lim_{x \rightarrow 0} \frac{\cos^2 x - \cos x - e^x \cos x + e^x - \frac{x^3}{2}}{x^n}$ is a finite nonzero number.

$$[\text{Ans. : } 4]$$

DIBY-16

If $\lim_{x \rightarrow 0} \frac{\cos 4x + a \cos 2x + b}{x^4}$ is finite, find a and b & also, find the limit.

$$[\text{Ans. : } a = -4, b = 3 \text{ \& Limit} = 8]$$

USE OF EXPANSION & FINDING UNKNOWN IF LIMIT IS GIVEN

DIBY-17

Let $\beta = \lim_{x \rightarrow 0} \frac{\alpha x - (e^{3x} - 1)}{\alpha x (e^{3x} - 1)}$ for some $\alpha \in R$. Then the value of $\alpha + \beta$ is: [JEE Mains-2022]

(A) $\frac{14}{5}$

(B) $\frac{3}{2}$

(C) $\frac{5}{2}$

(D) $\frac{7}{2}$

[Ans.: (C)]

DIBY-18

If $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 - x + 1} - ax \right) = b$, then the ordered pair (a, b) is: [JEE Mains-2021]

(A) $\left(-1, \frac{1}{2}\right)$

(B) $\left(-1, -\frac{1}{2}\right)$

(C) $\left(1, -\frac{1}{2}\right)$

(D) $\left(1, \frac{1}{2}\right)$

[Ans.: (C)]

DIBY-19

If $\lim_{n \rightarrow \infty} \left(\sqrt{n^2 - n - 1} + n\alpha + \beta \right) = 0$, then $8(\alpha + \beta)$ is equal to [JEE Mains-2022]

(A) 4

(B) -8

(C) -4

(D) 8

[Ans.: (C)]

USE OF EXPANSION & FINDING UNKNOWN IF LIMIT IS GIVEN



DIBY-20

If $\lim_{x \rightarrow 0} \frac{\alpha x e^x - \beta \log_e(1+x) + \gamma x^2 e^{-x}}{x \sin^2 x} = 10$, $\alpha, \beta, \gamma \in R$, then the value of

$\alpha + \beta + \gamma$ is _____.

[JEE Mains-2021]

[Ans. : 3]

DIBY-21

If $\lim_{x \rightarrow 0} \frac{a e^x - b \cos x + c e^{-x}}{x \sin x} = 2$, then $a + b + c$ is equal to

[JEE Mains-2021]

[Ans. : 4]

DIBY-22

If $\lim_{x \rightarrow \infty} \left(\frac{a(2x^3 - x^2) + b(x^3 + 5x^2 - 1) - c(3x^3 + x^2)}{a(5x^4 - x) - bx^4 + c(4x^4 + 1) + 2x^2 + 5x} \right) = 1$ and $a + b + c$ can be expressed as p/q in lowest form then find $p + q$.

[Ans. : 167]

USE OF EXPANSION & FINDING UNKNOWN IF LIMIT IS GIVEN



DIBY-23

Let $\alpha, \beta \in \mathbb{R}$ be such that $\lim_{x \rightarrow 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$. Then $6(\alpha + \beta)$ equals _____.

[JEE (Adv.)-2016]

[Ans. : 7]

DIBY-24

If $\lim_{x \rightarrow 0} \frac{e^{(a-1)x} + 2\cos bx + (c-2)e^{-x}}{x\cos x - \log_e(1+x)} = 2$, then $a^2 + b^2 + c^2$ is equal to

[JEE Mains-2026]

[Ans. : 7]

DIBY-25

If $\lim_{x \rightarrow 1} \frac{\sin(3x^2 - 4x + 1) - x^2 + 1}{2x^3 - 7x^2 + ax + b} = -2$, then the value of $(a - b)$ is equal to _____.

[JEE Mains-2022]

[Ans. : 11]

USE OF EXPANSION & FINDING UNKNOWN IF LIMIT IS GIVEN



DIBY-26

If $\lim_{x \rightarrow 0} \frac{e^{ax} - \cos(bx) - \frac{cxe^{-cx}}{2}}{1 - \cos(2x)} = 17$, then $5a^2 + b^2$ is

[JEE Mains-2023]

[Ans. : 68]

DIBY-27

For $\alpha, \beta, \gamma \in \mathbb{R}$, if $\lim_{x \rightarrow 0} \frac{x^2 \sin \alpha x + (\gamma - 1)e^{x^2}}{\sin 2x - \beta x} = 3$ then $\beta + \gamma - \alpha$ is equal to

[JEE Mains-2025]

[Ans. : 7]

DIBY-28

If $\lim_{x \rightarrow 1^+} \frac{(x-1)(6 + \lambda \cos(x-1)) + \mu \sin(1-x)}{(x-1)^3} = -1$, where $\lambda, \mu \in \mathbb{R}$, then $\lambda + \mu$ is equal to:

[JEE Mains-2025]

[Ans. : 18]

USE OF EXPANSION & FINDING UNKNOWN IF LIMIT IS GIVEN



DIBY-29

If the $\lim_{x \rightarrow 0} \frac{1}{x^3} \left(\frac{1}{\sqrt{1+x}} - \frac{1+ax}{1+bx} \right)$ exists and has the value equal to l , then find the value of $\frac{1}{8} \left(\frac{1}{a} - \frac{2}{l} + \frac{3}{b} \right)$.

[Ans. : 9]

DIBY-30

If $\lim_{x \rightarrow 0} \frac{\ln \cot\left(\frac{\pi}{4} - \beta x\right)}{\tan \alpha x} = 1$, then $\frac{\alpha}{\beta} = \underline{\hspace{2cm}}$

[Ans. : 2]

DIBY-31

If $\lim_{x \rightarrow 0} \frac{a \cot x}{x} + \frac{b}{x^2} = \frac{1}{3}$, then $b - a =$

[Ans. : 2]

USE OF EXPANSION & FINDING UNKNOWN IF LIMIT IS GIVEN



DIBY-32

$$\lim_{x \rightarrow 0} \frac{ax^2 e^x - b \log_e(1+x) + cxe^{-x}}{x^2 \sin x} = 1, \text{ then } 16(a^2 + b^2 + c^2) \text{ is equal to } \underline{\hspace{2cm}}.$$

[JEE Mains-2024]

[Ans. : 81]

DIBY-33

$$\text{If } \lim_{x \rightarrow 2} \frac{\sin(x^3 - 5x^2 + ax + b)}{(\sqrt{x-1} - 1) \log_e(x-1)} = m, \text{ then } a + b + m \text{ is equal to}$$

[JEE Mains-2026]

[Ans. : 6]

DIBY-34

The absolute value of product of all possible values of α , for which

$$\lim_{x \rightarrow 0} \left(\frac{1 - \cos(\alpha x) \cos((\alpha+1)x) \cos((\alpha+2)x)}{\sin^2((\alpha+1)x)} \right) = 2, \text{ is}$$

[JEE Mains-2026]

[Ans. : 1]

USE OF EXPANSION & FINDING UNKNOWN IF LIMIT IS GIVEN

DIBY-35

If $\lim_{x \rightarrow 0} \left(\frac{1 - \cos 3x \cdot \cos 9x \cdot \cos 27x \cdots \cos 3^n x}{1 - \cos \frac{x}{3} \cdot \cos \frac{x}{9} \cdot \cos \frac{x}{27} \cdots \cos \frac{x}{3^n}} \right) = 3^{10}$, then find the value of n .

[Ans. : 4]

DIBY-36

$\lim_{x \rightarrow \infty} \left(x - x^2 \ln \left(1 + \frac{1}{x} \right) \right)$ is equal to:

(A) $\frac{1}{2}$

(B) $\frac{3}{2}$

(C) $\frac{1}{3}$

(D) 1

[Ans. : (A)]

USE OF EXPANSION & FINDING UNKNOWN IF LIMIT IS GIVEN



DIBY-37

If $\alpha > \beta > 0$ are the roots of the equation $ax^2 + bx + 1 = 0$, and

$$\lim_{x \rightarrow \frac{1}{\alpha}} \left(\frac{1 - \cos(x^2 + bx + a)}{2(1 - \alpha x)^2} \right)^{\frac{1}{2}} = \frac{1}{k} \left(\frac{1}{\beta} - \frac{1}{\alpha} \right), \text{ then } k \text{ is equal to}$$

[JEE Mains-2023]

(A) 2β

(B) α

(C) 2α

(D) β

[Ans.: (c)]

DIBY-38

Let $\lim_{x \rightarrow 2} \frac{(\tan(x-2))(rx^2 + (p-2)x - 2p)}{(x-2)^2} = 5$ for some $r, p \in R$. If the set of all possible values of q , such that the roots of the equation $rx^2 - px + q = 0$ lie in $(0, 2)$, is the interval $(\alpha, \beta]$, then $4(\alpha + \beta)$ equals

[JEE Mains-2026]

[Ans.: 17]

DIBY-39

The value of the limit $\lim_{x \rightarrow \pi/2} \frac{4\sqrt{2}(\sin 3x + \sin x)}{(2\sin 2x \sin \frac{3x}{2} + \cos \frac{5x}{2}) - (\sqrt{2} + \sqrt{2}\cos 2x + \cos \frac{3x}{2})}$ is _____.

[JEE (Adv.)-2020]

[Ans. : 8]

DIBY-40

Let α be a positive real number. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : (\alpha, \infty) \rightarrow \mathbb{R}$ be the functions defined

by $f(x) = \sin\left(\frac{\pi x}{12}\right)$ and $g(x) = \frac{2\log_e(\sqrt{x} - \sqrt{\alpha})}{\log_e(e^{\sqrt{x}} - e^{\sqrt{\alpha}})}$. Then the value of $\lim_{x \rightarrow \alpha^+} f(g(x))$

is.

[JEE (Adv.)-2022]

[Ans. : 0.5]



DIBY-41

Solve : $\lim_{x \rightarrow \frac{\pi}{2}} \frac{a^{\cot x} - a^{\cos x}}{(\cot x - \cos x)}$

[Ans. : $\ln a$]

DIBY-42

If $\lim_{x \rightarrow 0^+} \frac{\tan\left(5(x)^{\frac{1}{3}}\right) \log_e(1+3x^2)}{(\tan^{-1} 3\sqrt{x})^2 (e^{5(x)^{\frac{4}{3}}} - 1)}$ is equal to:

[JEE Mains-2025]

(A) $\frac{1}{15}$

(B) 1

(C) $\frac{1}{3}$

(D) $\frac{5}{3}$

[Ans. : (C)]

DIBY-43

If $\Delta_r = \begin{vmatrix} r(r-1) & 2(r+2) & 2015 \\ r & 1 & -6 \\ r^2 & 4r & 0 \end{vmatrix}$ then $\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \Delta_r}{n^4}$ is equal to :

[Ans. : 3]

DIBY-44

If α and β , ($\alpha < \beta$) are the roots of the equation

$$\lim_{t \rightarrow \infty} \cos^{-1} \left[\sin \left(\tan^{-1} \left(\frac{\sqrt{tx}}{\sqrt{tx^2 - 3tx + t - 1 - x}} \right) \right) \right] = \frac{\pi}{6}, \text{ then the value of } (8^\alpha + 2^\beta - \alpha\beta).$$

[Ans. : 9]

DIBY-45

Let a sequence of number is as follows :

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 3 & & 5 & & \\ & 7 & & 9 & & 11 & \\ 13 & & 15 & & 17 & & 19 \\ 21 & 23 & 25 & 27 & 29 & & \end{array}$$

.....

If t_n is the first term of n^{th} row then $\lim_{n \rightarrow \infty} (\sqrt{t_n} - n)$ is equal to :

[Ans. : -1/2]

DIBY-46



Evaluate : $\lim_{k \rightarrow 0} \left(\frac{\ln(\ln(1+k^4))}{\ln(\ln(1+k^2))} \right)$

[Ans. : 2]

DIBY-47

$$L = \lim_{x \rightarrow 0} \frac{\sqrt{\frac{\cos 2x + (1+3x)^{1/3}}{2}} - \sqrt[3]{\frac{4\cos^3 x - \ln(1+x)^4}{4}}}{x}$$

If $L = a/b$ where ' a ' and ' b ' are relatively primes find $(a + b)$.

[Ans. : 19]

DIBY-48

Find $\lim_{x \rightarrow 0} \left(\frac{\tan^3 x - \tan x^3}{x^5} \right)$.

[Ans. : 1]

DIBY-49

If the equation of tangent drawn to the curve $y = f(x)$, at the point $(3, 5)$ is

$$5x - 4y + 5 = 0 \text{ and } \lim_{x \rightarrow 3} \frac{(3^{4f(x)} - 2(1 + 3 + 3^2 + \dots + 3^{19}) - 1)^2}{1 - \cos(\log_e(4 - x))} = 2(a \cdot 3^b \cdot \log_e c)^2, \text{ where}$$

a and c are prime numbers and $b \in N$, then $\frac{(a + b + c)}{10}$ equals

[Ans. : 2.8]

DIBY-50 (Paragraph)

Let $f(x)$ be a polynomial satisfying

$$\lim_{x \rightarrow \infty} \frac{x^4 f(x)}{x^8 + 1} = 3, f(2) = 5, f(3) = 10, f(-1) = 2 \text{ and } f(-6) = 37.$$

(A) The value of $f(0)$ is equal to:

[Ans. : 109]

(B) The value of $\lim_{x \rightarrow -6} \frac{f(x) - x^2 - 1}{3(x + 6)}$, is equal to :

[Ans. : -360]

DIBY-51 (Paragraph)

Let $P(x)$ be a polynomial of degree '6' with leading coefficient unity and

$$P(-x) = P(x) \quad \forall \quad x \in \mathbf{R}. \text{ Also } (P(1) + 3)^2 + P^2(2) + (P(3) - 5)^2 = 0$$

(A) The value of $\lim_{x \rightarrow -2} \frac{\sin(P(x))}{(x-2) \tan(x+2)}$ is equal to :

[Ans. : -14]

(B) The value of $\lim_{x \rightarrow \infty} \frac{\left(x^2 - \sqrt{\frac{P(x)}{x^2 - 4}}\right)}{x \tan \frac{1}{x}}$ is equal to :

[Ans. : 5]

DIBY-52 (Built in Limit)

Let $f(x) = \lim_{n \rightarrow \infty} \frac{x^2 + 2(x+1)^{2n}}{(x+1)^{2n+1} + x^2 + 1}$, $n \in \mathbf{N}$ and $g(x) = \tan \left(\frac{1}{2} \sin^{-1} \left\{ \frac{2f(x)}{1+f^2(x)} \right\} \right)$. If $x \in (-2, 0)$

then range of $g(x)$ is

[Ans. : $\left(0, \frac{4}{5}\right)]$

EXPONENTIAL LIMITS

DIBY-53

If a, b, c, d are positive, then $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{a+bx}\right)^{c+dx}$

(A) $e^{d/b}$

(B) $e^{c/a}$

(C) $e^{(c+d)/(a+b)}$

(D) e

[Ans.: (A)]

DIBY-54

If $\lim_{x \rightarrow 0} \left(1 + x \ln(1 + b^2)\right)^{\frac{1}{x}} = 2b \sin^2 \theta, b > 0$ and $\theta \in (-\pi, \pi)$, then the value of θ is

(A) $\pm \frac{\pi}{4}$

(B) $\pm \frac{\pi}{3}$

(C) $\pm \frac{\pi}{6}$

(D) $\pm \frac{\pi}{2}$

[Ans.: (D)]

DIBY-55

If $\lim_{x \rightarrow \infty} \left(\left(\frac{e}{1-e}\right) \left(\frac{1}{e} - \frac{x}{1+x}\right)\right)^x = \alpha$, then the value of $\frac{\log_e \alpha}{1 + \log_e \alpha}$ equals

[JEE Mains-2025]

(A) e^{-1}

(B) e^2

(C) e^{-2}

(D) e

[Ans.: (D)]



EXPONENTIAL LIMITS

DIBY-56

Let $p = \lim_{x \rightarrow 0^+} (1 + \tan^2 \sqrt{x})^{\frac{1}{2x}}$ then $\log p$ is equal to

- (A) $\frac{1}{2}$
- (B) $\frac{1}{4}$
- (C) 2
- (D) 0

[Ans. : (A)]

DIBY-57 (Match the Column)

Column I	Column II
(A) $\lim_{x \rightarrow \infty} \left(\frac{x}{1+x}\right)^x$ equals	(P) e^2
(B) $\lim_{x \rightarrow \infty} \left(\sin \frac{1}{x} + \cos \frac{1}{x}\right)^x$	(Q) $e^{-1/2}$
(C) $\lim_{x \rightarrow 0} (\cos x)^{\cot^2 x}$	(R) E
(D) $\lim_{x \rightarrow 0} \left(\tan \left(\frac{\pi}{4} + x\right)\right)^{1/x}$	(S) e^{-1}

[Ans. : (A)-S; (B)-R; (C)-Q; (D)-P]



DIBY-58

EXPONENTIAL LIMITS

If $\alpha = \lim_{x \rightarrow 0^+} \left(\frac{e^{\sqrt{\tan x}} - e^{\sqrt{x}}}{\sqrt{\tan x} - \sqrt{x}} \right)$ and $\beta = \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{2} \cot x}$ are the roots of the quadratic equation $ax^2 + bx - \sqrt{e} = 0$, then $12 \log_e(a + b)$ is equal to _____. [JEE Mains-2024]
[Ans.: 6]

DIBY-59

$\lim_{x \rightarrow 0} \frac{e - (1 + 2x)^{\frac{1}{2x}}}{x}$ is equal to

[JEE Mains-2024]

(A) 0

(B) $-\frac{2}{e}$

(C) e

(D) $e - e^2$

[Ans.: (C)]

DIBY-60

$\lim_{x \rightarrow 0} \left(\frac{(1+x)^{\frac{2}{x}}}{e^2} \right)^{\frac{4}{\sin x}}$ is

[Ans.: e^{-4}]

EXPONENTIAL LIMITS



DIBY-61

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function satisfying $f'(3) + f'(2) = 0$. Then

$$\lim_{x \rightarrow 0} \left(\frac{1 + f(3+x) - f(3)}{1 + f(2-x) - f(2)} \right)^{\frac{1}{x}} \text{ is}$$

[JEE Mains-2019]

[Ans. : 1]

DIBY-62

If $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}} = p$, then $96 \log_e p$ is equal to _____.

[JEE Mains-2025]

[Ans. : 32]

DIBY-63

Let $f: \mathbb{R} \rightarrow (0, \infty)$ be a twice differentiable function such that $f(3) = 18, f'(3) = 0$

and $f''(3) = 4$. Then $\lim_{x \rightarrow 1} \left(\log_e \left(\frac{f(2+x)}{f(3)} \right)^{\frac{18}{(x-1)^2}} \right)$ is equal to

[JEE Mains-2026]

[Ans. : 2]

EXPONENTIAL LIMITS

DIBY-64

Let $P(x) = 100x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \forall a_i \in R; 0 \leq i \leq n-1$.

Let $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ be its n roots such that $\prod_{i=2}^n (\alpha_1 - \alpha_i) = K, (K \in R - \{0\})$.

If $L = \lim_{x \rightarrow \alpha_1} (1 + P(x))^{\frac{1}{x - \alpha_1}}$ where α_1 is a real root, then find the value of $\frac{\ln L}{K}$.

[Ans. : 100]

DIBY-65

$\lim_{x \rightarrow 0} \left(1 + \frac{a \sin bx}{\cos x}\right)^{\frac{1}{x}}$, where a, b are non-zero constants is equal to :

[Ans. : e^{ab}]

DIBY-66

$\lim_{x \rightarrow \infty} \left(\frac{\cosh(\pi/x)}{\cos(\pi/x)}\right)^{x^2}$ where $\cosh t = \frac{e^t + e^{-t}}{2}$

[Ans. : $((e)^{\pi^2})$]

EXPONENTIAL LIMITS

DIBY-67 (Paragraph)

Let $f(x) = \lim_{n \rightarrow \infty} \left(\cos \sqrt{\frac{x}{n}} \right)^n$, $g(x) = \lim_{n \rightarrow \infty} \left(1 + x(-1 + e^{1/n}) \right)^n$ and

$$h(x) = \tan^{-1} \left(g^{-1}(f^{-1}(x)) \right).$$

(A) Value of $\lim_{x \rightarrow 0} \frac{\ln(f(x))}{\ln(g(x))}$ is equal to

[Ans. : -1/2]

(B) Domain of the function $h(x)$ is

[Ans. : (0, 1)]

(C) Range of the function $h(x)$ is

[Ans. : $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$]

EXPONENTIAL LIMITS

DIBY-68

The value of $\lim_{x \rightarrow 0} \left((\cos x)^{\frac{1}{\sin^2 x}} + \frac{\sin 2x + 2 \tan^{-1} 3x + 3x^2}{\ln(1 + 3x + \sin^2 x) + xe^x} \right)$ is :

[Ans. : $\frac{1}{\sqrt{e}} + 2$]

DIBY-69

If $\lim_{x \rightarrow 0} \frac{f(x)}{\sin^2 x} = 8$, $\lim_{x \rightarrow 0} \frac{g(x)}{2 \cos x - xe^x + x^3 + x - 2} = \lambda$ and $\lim_{x \rightarrow 0} (1 + 2f(x))^{\frac{1}{g(x)}} = \frac{1}{e}$,

then $\lambda =$ _____.

[Ans. : 8]

LIMITS USING T.M.O.D.



DIBY-70

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $f(0) = 1$ and $f(2x) - f(x) = x$ for all $x \in \mathbb{R}$. If $\lim_{n \rightarrow \infty} \left\{ f(x) - f\left(\frac{x}{2^n}\right) \right\} = G(x)$, then $\sum_{r=1}^{10} G(r^2)$ is equal to **[JEE Mains-2025]**

- (A) 540 (B) 385 (C) 420 (D) 215

[Ans.: (B)]

DIBY-71

The value of $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} \right)$ is:

[JEE Mains-2025]

- (A) $\frac{5}{3}$ (B) $\frac{4}{3}$ (C) $\frac{7}{3}$ (D) 2

[Ans.: (A)]

DIBY-72

$\lim_{x \rightarrow 2} \left(\sum_{n=1}^9 \frac{x}{n(n+1)x^2 + 2(2n+1)x + 4} \right)$ is equal to:

[JEE Mains-2021]

- (A) $\frac{9}{44}$ (B) $\frac{5}{24}$ (C) $\frac{7}{36}$ (D) $\frac{1}{5}$

[Ans. (A)]

SANDWICH THEOREM



DIBY-73

Let $[.]$ denote the greatest integer function and $f(x) = \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n \left[\frac{k^2}{3^x} \right]$. Then

$12 \sum_{j=1}^{\infty} f(j)$ is equal to

[JEE Mains-2026]

[Ans. : 2]

DIBY-74

Evaluate : $\lim_{n \rightarrow \infty} \left(\frac{n}{n^2+1} + \frac{n}{n^2+2} + \frac{n}{n^2+3} + \dots + \frac{n}{n^2+n} \right)$.

[Ans. : 1/2]

DIBY-75

Find the value of $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \dots + \frac{1}{\sqrt{n^2+2n}}$.

[Ans. : 2]

SANDWICH THEOREM

DIBY-76

Evaluate $\lim_{n \rightarrow \infty} \left(\frac{n}{n^2+1} + \frac{n}{n^2+2} + \frac{n}{n^2+3} + \dots + \frac{n}{n^2+n} \right).$

[Ans. : 1/2]

DIBY-77

Evaluate $\lim_{n \rightarrow \infty} \left(\frac{1}{n^3+1} + \frac{4}{n^3+4} + \frac{9}{n^3+9} + \dots + \frac{n^2}{n^3+n^2} \right).$

[Ans. : 1/3]

DIBY-78

If $f: (0, \infty) \rightarrow \mathbb{N}$ and

$f(x) = \left[\frac{x^2+x+1}{x^2+1} \right] + \left[\frac{4x^2+x+2}{2x^2+1} \right] + \left[\frac{9x^2+x+3}{3x^2+1} \right] + \dots + \left[\frac{n^2x^2+x+n}{nx^2+1} \right], n \in \mathbb{N}$ then find the value of

$\lim_{n \rightarrow \infty} \left[\frac{f(x)-n}{(f(x))^2 - \frac{n^3(n+2)}{4}} \right].$

[Note : $[t]$ denotes the greatest integer less than or equal to t .

[Ans. : 2]

LIMITS INVOLVING GIF, FPF & MOD



DIBY-79

Let $[x]$ denote the greatest integer less than or equal to x . Then

$$\lim_{x \rightarrow 0} \frac{\tan(\pi \sin^2 x) + (|x| - \sin(x[x]))^2}{x^2}$$

(A) Does not exist

(C) Equals $\pi + 1$

(B) Equals π

(D) Equals 0

[JEE Mains-2019]

[Ans.: (A)]

DIBY-80

Let $[t]$ denote the greatest integer function $\leq t$. If for some $\lambda \in R - \{0, 1\}$,

$$\lim_{x \rightarrow 0} \left| \frac{1-x+|x|}{\lambda-x+[x]} \right| = L, \text{ then } L \text{ is equal to}$$

(A) 2

(B) $\frac{1}{2}$

(C) 0

(D) 1

[JEE Mains-2020]

[Ans.: (A)]

DIBY-81

$$\lim_{x \rightarrow 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2}$$

(A) is equal to -1

(C) is equal to 1

(B) does not exist

(D) is equal to 2

[JEE Mains-2024]

[Ans.: (D)]

LIMITS INVOLVING GIF, FPF & MOD

DIBY-82

For each $x \in R$, let $[x]$ be the greatest integer less than or equal to x .

Then $\lim_{x \rightarrow 0^-} \frac{x([x] + |x|)\sin[x]}{|x|}$ is equal to

[JEE Mains-2019]

- (A) $-\sin 1$ (B) 0 (C) 1 (D) $\sin 1$

[Ans.: (A)]

DIBY-83

Let $f(x) = \begin{cases} x - 1, & x \text{ is even} \\ 2x, & x \text{ is odd} \end{cases}, x \in N$. If for some $a \in N$, $f(f(f(a))) = 21$,

then $\lim_{x \rightarrow a^-} \left\{ \frac{|x|^3}{a} - \left[\frac{x}{a} \right] \right\}$, where $[t]$ represents GIF, is equal to:

[JEE Mains-2024]

- (A) 121 (B) 144 (C) 169 (D) 225

[Ans.: (B)]

DIBY-84

$\lim_{n \rightarrow \infty} \left(\frac{\sin\left\{\frac{2}{n}\right\}}{\left[2n \tan \frac{1}{n}\right] \left(\tan \frac{1}{n}\right)} + \frac{1}{n^2 + \cos n} \right)^{n^2}$ Where $[] \rightarrow \text{G.I.F.}$ and $\{ \}$ is FPF

[Ans.: 1]

MISCLELLANEOUS QUESTIONS

DIBY-85

If the function $f(x) = \frac{\tan(\tan x) - \sin(\sin x)}{\tan x - \sin x}$ is continuous at $x = 0$, then $f(0)$ is equal to

[JEE Mains-2025]

[Ans. : 2]

DIBY-86

For some (a, b) , let $f(x) = \begin{vmatrix} a + \frac{\sin x}{x} & 1 & b \\ a & 1 + \frac{\sin x}{x} & b \\ a & 1 & b + \frac{\sin x}{x} \end{vmatrix}$ $x \neq 0$, $\lim_{x \rightarrow 0} f(x) = \lambda + \mu a + \nu b$.

Then $(\lambda + \mu + \nu)^2$ is equal to :

(A) 25

(B) 9

(C) 16

(D) 36

[JEE Mains-2025]

[Ans. : (c)]

MISCELLANEOUS QUESTIONS



DIBY-87

$\lim_{t \rightarrow 0} \left(1^{\frac{1}{\sin^2 t}} + 2^{\frac{1}{\sin^2 t}} + \dots + n^{\frac{1}{\sin^2 t}} \right)^{\sin^2 t}$ is equal to

[JEE Mains-2023]

- (A) $n^2 + n$ (B) n (C) $\frac{n(n+1)}{2}$ (D) n^2

[Ans.: (B)]

DIBY-88

Let $f(x)$ be a polynomial of degree 6 in x , in which the coefficient of x^6 is unity and it has **extrema** at $x = -1$ and $x = 1$. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^3} = 1$ then $5f(2)$ is equal to ____.

[JEE Mains-2021]

[Ans.: 144]

DIBY-89

Let $x_1 = 1$ and $x_n = \sqrt{2 + x_{n-1}}$, $n = 2, 3, \dots$, $\lim_{n \rightarrow \infty} x_n$

[Ans.: 2]



MISCELLANEOUS QUESTIONS

DIBY-90

If $a_{n+1} = a_n + \sqrt{1 + a_n^2}$ and $a_0 = 0$, find $\lim_{n \rightarrow \infty} \left(\frac{a_n}{2^{n-1}} \right)$.

[Ans. : $4/\pi$]

DIBY-91

If $S = \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}} \cdots \cdots \cdots \infty$, then find the value of πS .

[Ans. : 2]

DIBY-92

Let O be the center of a circle of radius one. P, Q are the points on the circle such that $\theta = \angle POQ$ is an acute angle and R is a point outside the circle such that $OPRQ$ is a parallelogram. If the area of the part of the parallelogram that is outside the circle is $f(\theta)$, then $\lim_{\theta \rightarrow 0} \frac{f(\theta)}{\theta}$ is equal to :

[Ans. : $1/2$]

MISCELLANEOUS QUESTIONS

DIBY-93

Let AB be a fixed diameter of a circle of unit radius and P be a variable point on this circle. Let M be foot of the perpendicular drawn from the point P to the

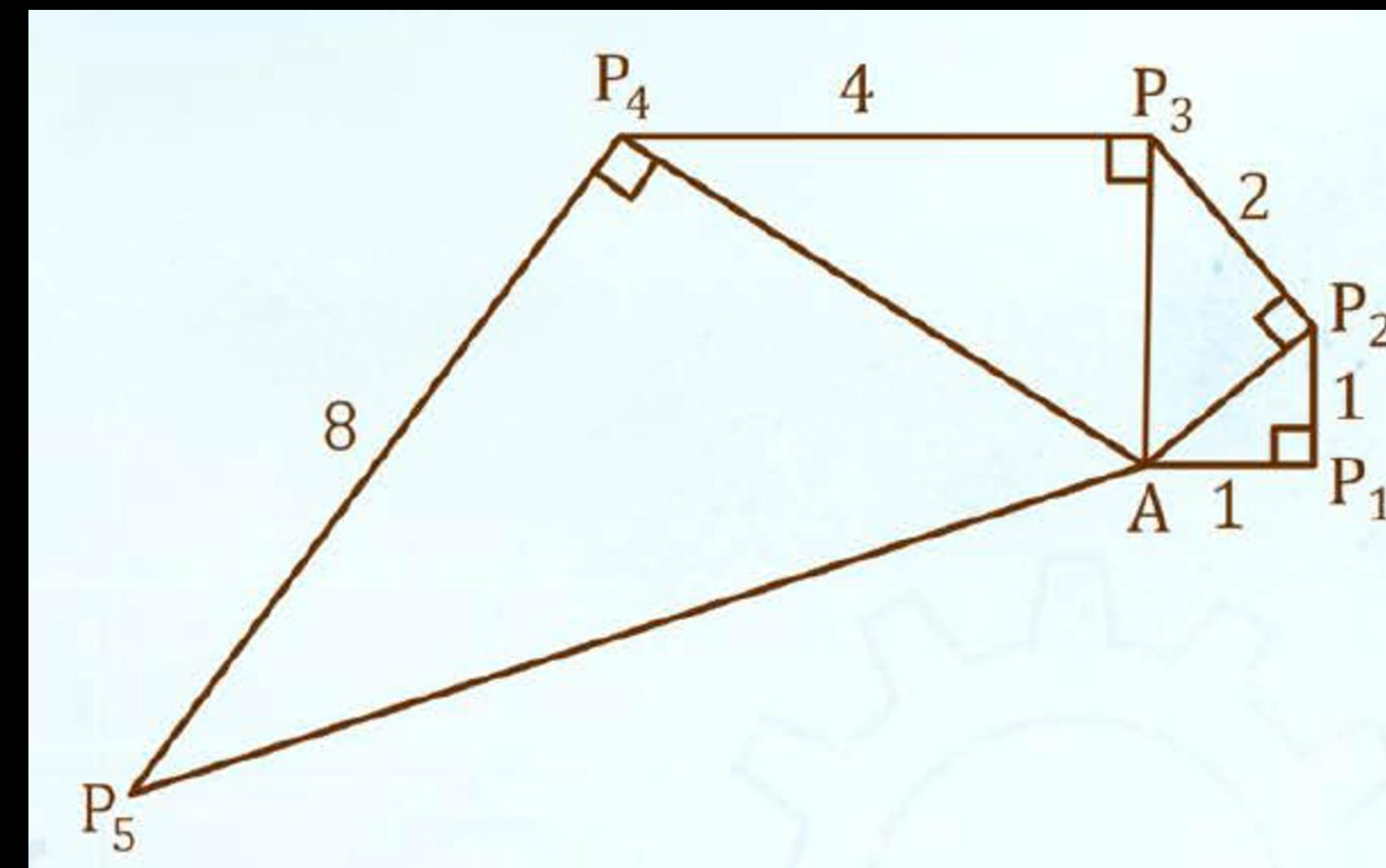
tangent at point A , then $\lim_{p \rightarrow A} \frac{\text{Area of } \triangle PAM}{(\text{Minor arc length } PA)^3}$ is equal to [Ans. : (1/4)]

DIBY-94

Let $\{P_n\}$ be a sequence of points determined as in the figure.

Thus $|AP_1| = 1$, $|P_n P_{n+1}| = 2^{n-1}$, and angle $AP_n P_{n+1}$ is a right angle. $\lim_{n \rightarrow \infty} \angle P_n A P_{n+1}$ equals:

- (A) $\frac{\pi}{6}$
- (B) $\frac{\pi}{4}$
- (C) $\frac{\pi}{3}$
- (D) $\frac{5\pi}{12}$



[Ans. : (c)]

DIBY-95

MISCLELLANEOUS QUESTIONS



Evaluate: $\lim_{x \rightarrow \infty} \left\{ \frac{x}{x + \frac{3\sqrt{x}}{x + \frac{3\sqrt{x}}{\dots \text{infinity}}}} \right\}$

[Ans. : 1]

DIBY-96

$$\lim_{x \rightarrow \infty} \left(\frac{1^{1/x} + 2^{1/x} + 3^{1/x} + \dots + n^{1/x}}{n} \right)^{nx}.$$

[Ans. : $(n!)$]

DIBY-97

Let $f(x) = \begin{cases} x + 1, & x > 0 \\ 2 - x, & x \leq 0 \end{cases}$ and $g(x) = \begin{cases} x + 3, & x < 1 \\ x^2 - 2x - 2, & 1 \leq x < 2 \\ x - 5, & x \geq 2 \end{cases}$. Find L.H.L. and R.H.L.

of $g(f(x))$ at $x = 0$ and hence find $\lim_{x \rightarrow 0} g(f(x))$

[Ans. : -3]

MISCELLANEOUS QUESTIONS



DIBY-98

If α is the positive root of the equation, $p(x) = x^2 - x - 2 = 0$ then $\lim_{x \rightarrow \alpha^+} \frac{\sqrt{1 - \cos(p(x))}}{x + \alpha - 4}$ is equal to: [JEE Mains-2020]

(A) $1/\sqrt{2}$

(B) $1/2$

(C) $3/\sqrt{2}$

(D) $3/2$

[Ans.: (C)]

DIBY-99

If α, β are two distinct real roots of the equation $ax^3 + x - 1 - a = 0, (a \neq -1, 0)$

none of which is equal to unity, then the value of $\lim_{x \rightarrow (1/\alpha)} \frac{(1+a)x^3 - x^2 - a}{(e^{1-\alpha x} - 1)(x-1)}$ is $\frac{al(k\alpha - \beta)}{\alpha}$.

Find the value of kl .

[Ans.: 1]

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians